

Phase 9 Closeout Budget-Control CVRP Aggregate Report

Overview

- Run count: 20.
- Condition count: 6.
- Best held-out condition: phase9_closeout_baseline_clarke_wright_savings.

Condition Means

Condition	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity	Robustness Dispersion
phase9_closeout_baseline_clarke_wright_savings	1.0	0.073766	0.073766	78.329757	0.0	0.0	0.0	0.0	0.0	0.007791
phase9_closeout_baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	43.786892	0.0	0.0	0.0	0.0	0.0	0.002777
phase9_closeout_baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1439.817186	0.0	0.0	0.0	0.0	0.0	0.086899
phase9_closeout_budget_matched_no_replay	0.15	2.932746	0.382746	229.661432	1.5	0.0	0.0	0.955648	13.80875	0.10789
phase9_closeout_direct_generate_plus_one_repair	0.25	5.286402	0.018402	275.998501	1.35	0.0	0.0	0.922332	14.901	0.485845
phase9_closeout_replay_solver_evolution	1.0	0.171114	0.171114	446.706369	1.0	0.0	0.0	0.390074	7.344867	0.003097

Paired Comparisons

savings_vs_nearest_neighbor

- Candidate: phase9_closeout_baseline_clarke_wright_savings versus reference phase9_closeout_baseline_nearest_neighbor_constructive across 20 paired runs.
- Held-out penalized-gap delta: -0.199669 (95% CI -0.199669 to -0.199669).
- Held-out feasibility-rate delta: 0.0 (95% CI 0.0 to 0.0).

insertion_vs_savings

- Candidate: phase9_closeout_baseline_regret_insertion_local_search versus reference phase9_closeout_baseline_clarke_wright_savings across 20 paired runs.
- Held-out penalized-gap delta: 0.392625 (95% CI 0.392625 to 0.392625).
- Held-out feasibility-rate delta: 0.0 (95% CI 0.0 to 0.0).

direct_vs_clarke_wright

- Candidate: phase9_closeout_direct_generate_plus_one_repair versus reference phase9_closeout_baseline_clarke_wright_savings across 20 paired runs.
- Held-out penalized-gap delta: 5.212636 (95% CI 3.054197 to 7.517502).
- Held-out feasibility-rate delta: -0.75 (95% CI -0.9 to -0.55).

budget_matched_vs_clarke_wright

- Candidate: phase9_closeout_budget_matched_no_replay versus reference phase9_closeout_baseline_clarke_wright_savings across 20 paired runs.
- Held-out penalized-gap delta: 2.85898 (95% CI 2.3404 to 3.456763).
- Held-out feasibility-rate delta: -0.85 (95% CI -1.0 to -0.7).

replay_vs_clarke_wright

- Candidate: phase9_closeout_replay_solver_evolution versus reference phase9_closeout_baseline_clarke_wright_savings across 20 paired runs.
- Held-out penalized-gap delta: 0.097348 (95% CI 0.055994 to 0.138558).
- Held-out feasibility-rate delta: 0.0 (95% CI 0.0 to 0.0).

direct_vs_budget_matched

- Candidate: phase9_closeout_direct_generate_plus_one_repair versus reference phase9_closeout_budget_matched_no_replay across 20 paired runs.
- Held-out penalized-gap delta: 2.353656 (95% CI 0.113058 to 4.737403).
- Held-out feasibility-rate delta: 0.1 (95% CI -0.15 to 0.35).

replay_vs_budget_matched

- Candidate: phase9_closeout_replay_solver_evolution versus reference phase9_closeout_budget_matched_no_replay across 20 paired runs.
- Held-out penalized-gap delta: -2.761632 (95% CI -3.369777 to -2.201586).
- Held-out feasibility-rate delta: 0.85 (95% CI 0.7 to 1.0).

replay_vs_direct

- Candidate: phase9_closeout_replay_solver_evolution versus reference phase9_closeout_direct_generate_plus_one_repair across 20 paired runs.
- Held-out penalized-gap delta: -5.115288 (95% CI -7.451608 to -2.946446).
- Held-out feasibility-rate delta: 0.75 (95% CI 0.55 to 0.9).